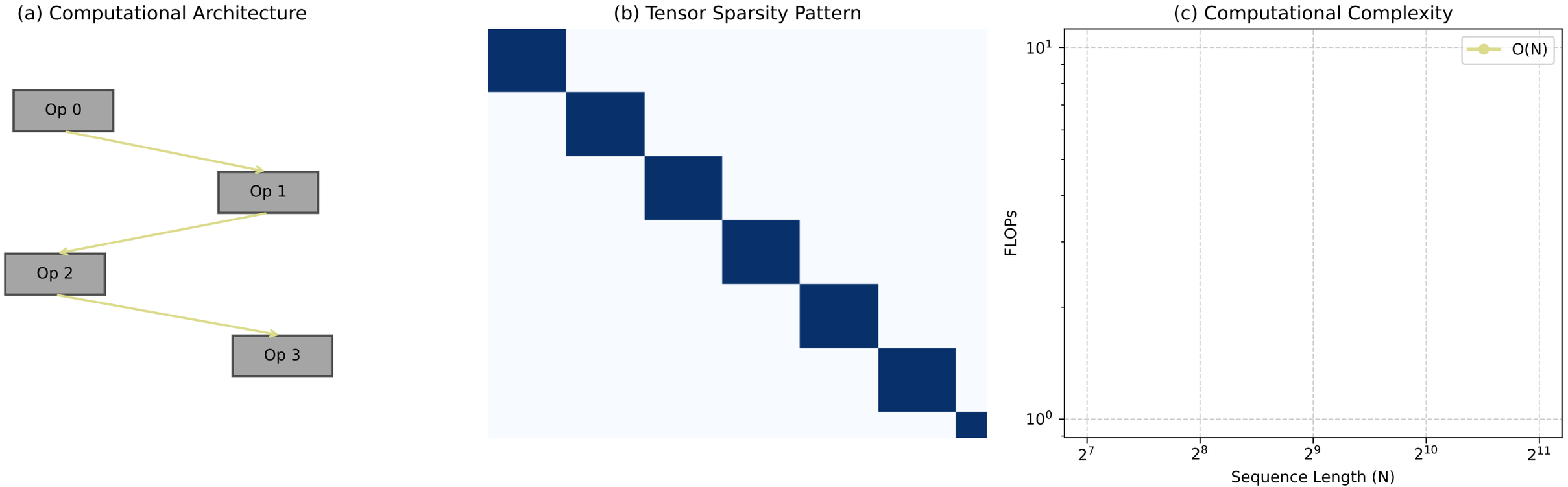Module Architecture: residuals